

Designing efficient urban bike path networks that meet the needs of cyclists

Designing efficient bike path networks requires balancing multiple opposing constraints such as cost and safety. An adaptive demand-driven inverse percolation approach is proposed to generate efficient network structures by explicitly taking into account the demands of cyclists and their route choice behavior based on safety preferences.

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The problem

Sufficiently developed infrastructure is needed to facilitate more sustainable transport and is of particular importance for promoting cycling over motorized individual transport, for example for inner-city trips. However, designing infrastructure networks constitutes a highly complex problem since the networks are subject to multiple, often opposing constraints. Bike path networks, for instance, should enable direct, fast, and safe travel without requiring every street to have a bike path. Advances in the theoretical study of bike path networks employ concepts from network percolation to understand the structure of the networks by focusing on individual perspectives such as the connectivity of the network¹ or the cycling demand². Previous studies successfully employed adaptive network percolation (that is, re-evaluating the order in which connections are added or removed after each step instead of following a predetermined order) to explain structural properties of other transportation networks³. In this study we wanted to explore whether we could combine these ideas to design even better bike path network structures.

The solution

The complexity of the network design problem originates from the interaction between the routes that cyclists choose to take and the available infrastructure. For example, cyclists do not always take the shortest routes, but may prefer longer trips to avoid cycling on streets with heavy car traffic without a bike path⁴. Therefore, we defined an effective network in which the lengths of street segments were modified based on the amount of car traffic. Representing the route choice behavior as an effective shortest path model on this network enabled us to integrate it into an adaptive inverse percolation framework. Instead of adding bike paths, we started from a perfect network where every street has a bike path and removed the least important bike paths step-by-step. After each step, we updated the routes of cyclists in response to the new network, keeping the network adapted to the current bike path usage. We repeated this process until all bike paths had been removed, thereby creating a sequence of bike path networks with different total lengths.

The input data required for our approach include the origin–destination

demand of cyclists, street network, and penalties for the route choice model. By combining bike sharing demand data and OpenStreetMap networks, we tested this approach for different cities. We found that a high overall 'bikeability' of a city's street network could be achieved by equipping a small fraction of streets with a bike path. Comparing the network generated by our algorithm with a network that equips all major streets with bike paths (Fig. 1) clearly shows that the network generated by our algorithm more accurately reflects the demand structure and results in a more bicycle-friendly network using the same total length of bike paths.

Future directions

Our approach relies on the available input data. As the networks are generated by considering the route choice model and demand, the resulting networks must be interpreted in this context. The route choice model currently only reflects the aversion to cycling on busy streets without a bike path but neglects other relevant factors for cyclists' route choices. Therefore, the generated networks could be improved by updating the route choice model to include additional deterrents, such as slopes and turns at junctions not controlled by traffic lights. Moreover, measured demand for bike paths and data on cyclists' route choices are likely to be strongly influenced by the bike paths that currently do or do not exist or the types of cyclists that are studied. Alternatively, modeling bike path networks to account for the desired or anticipated demand patterns, may help to future-proof bike path networks. Additionally, incorporating empirical accident data⁵ may enable a more direct evaluation of the safety of a proposed bike path network.

We hope that the ability to quickly generate bike path networks will enable a more in-depth analysis of common patterns and help to design guidelines for efficient bike path networks. Building on increasingly available detailed input data² and extensions of the framework to include the relevant economic and city planning constraints, the framework could eventually be used to support the planning of bike path networks across cities.

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EXPERT OPINION

"I think that this study is an important addition to the quantitative bicycle network literature, which is extremely scarce so far. The paper applies tools from network science and physics to an urban planning problem. I find the subtractive method beautiful

and elegantly simple. It is also a very nice complement and novelty compared to the few existing comparable approaches, and it could be a superior logical approach."

Michael Szell, IT University of Copenhagen, Copenhagen, Denmark

FIGURE

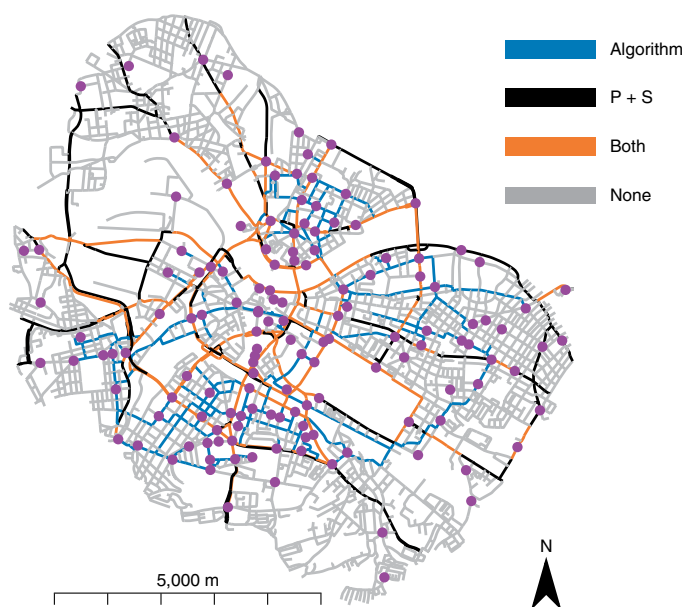


Fig. 1 | Demand-efficient bike path network for Dresden. Comparison between the network generated by our algorithm (blue edges) and the network where only major (primary and secondary) streets are equipped with a bike path (black edges). The grey edges indicate streets with no bike path. The orange edges show streets on which bike paths are proposed by both networks. Purple dots show the location of bike sharing stations used as demand input. There is good agreement between the two models but our algorithm more accurately reflects the input demand structure, and also keeps highly used smaller streets equipped with bike paths and exhibits a higher density of bike paths in high-demand areas, compared with the primary-secondary network. © 2022, Steinacker, C. et al.

BEHIND THE PAPER

This research idea emerged — as so many projects do — from a fortunate coincidence, with the right discussion occurring at the right time. As many members of our research group come to work by bike, the state of the cycling infrastructure in Dresden is one of the typical topics discussed over lunch. Unfortunately, this state is (in contrast to Dresden's public transportation) less than ideal in many places. When this discussion

coincided with the presentation of a paper¹ that introduced the possibility of using percolation to understand bike path networks, the idea quickly evolved into the dynamic percolation approach explained above. However, despite the topic being related to networks and mobility, getting into the new field of bike path networks proved to be a complex but also highly interesting and rewarding endeavor. **M.S.**

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FROM THE EDITOR

"The computational framework presented by Christoph Steinacker et al. here is an important contribution for the design of efficient bike path networks. My hope is that it will be further applied to quantitatively evaluate and design bike path network structures that balance both the practical constraints and cycling demand for an improved urban bikeability." **Jie Pan, Associate Editor, Nature Computational Science.**